$$\left[\textbf{Ans.}: \tan^{-1}\left(\frac{\sin^2\alpha}{1+\sin\alpha\cos\alpha}\right)\right]$$

7. $x\sin\alpha-\frac{1}{2}x^2\sin 2\alpha+\frac{1}{3}x^3\sin 3\alpha - \ldots \infty.$

$$\left[\textbf{Ans.}: \tan^{-1}\left(\frac{x\sin\alpha}{1+x\cos\alpha}\right)\right]$$

8. $e^{\alpha}\cos\beta-\frac{1}{3}e^{3\alpha}\cos 3\beta+\frac{1}{5}e^{5\alpha}\cos^5\beta + \ldots \infty.$

$$\left[\textbf{Ans.}: \frac{1}{2}\tan^{-1}\frac{\cos\beta}{\sinh\alpha}\right]$$

9. $\cos 2\alpha+\frac{1}{3}\cos 6\alpha+\frac{1}{5}\cos 10\alpha+\ldots\infty.$

[**Ans.** : $\log\cot\theta$]

10. $1-\frac{1}{2}\cos\alpha+\frac{1\cdot3}{2\cdot4}\cos 2\alpha-\frac{1\cdot3\cdot5}{2\cdot4\cdot6}\cos 3\alpha+\ldots\ldots\infty.$

$$\left[\textbf{Ans.}: (2\cos\alpha)^{-\frac{1}{2}}\cos\frac{\alpha}{4}\right]$$

11. $\frac{1}{2}\sin\alpha+\frac{1\cdot3}{2\cdot4}\sin 2\alpha+\frac{1\cdot3\cdot5}{2\cdot4\cdot6}\sin 3\alpha + \ldots\ldots (\alpha\neq n\pi).$

$$\left[\textbf{Ans.}: \left(2\sin\frac{\alpha}{2}\right)^{-\frac{1}{2}}\sin\left(\frac{\pi}{4}-\frac{\alpha}{4}\right)\right]$$

1.10 CIRCULAR AND HYPERBOLIC FUNCTIONS

1.10.1 Circular Function

From Euler's formula, we have

$$e^{i\theta}=\cos\theta+i\sin\theta$$
$$e^{-i\theta}=\cos\theta-i\sin\theta$$
$$\cos\theta=\frac{e^{i\theta}+e^{-i\theta}}{2},\ \sin\theta=\frac{e^{i\theta}-e^{-i\theta}}{2!}$$

If $z=x+iy$ is a complex number, then

$$\cos z=\frac{e^{iz}+e^{-iz}}{2},\ \sin z=\frac{e^{iz}-e^{-iz}}{2i}$$

These are called circular functions of complex numbers.

1.10.2 Hyperbolic Function

If z is a complex number, then sine hyperbolic of z is denoted by $\sinh z$ and is given as,

$$\sinh z=\frac{e^{z}-e^{-z}}{2}$$

and cosine hyperbolic of z is denoted by $\cosh z$ and is given as,

$$\cosh z=\frac{e^{z}+e^{-z}}{2}$$

From these expressions, other hyperbolic functions can also be obtained as

$$\tanh z=\frac{\sinh z}{\cosh z},\ \text{cosech } z=\frac{1}{\sinh z},\ \text{sech } z=\frac{1}{\cosh z}\ \text{and}\ \coth z=\frac{1}{\tanh z}.$$

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From the above definitions of sinh z, cosh z, tanh z, we can obtain the following val of hyperbolic functions.

z	$-\infty$	0	∞
sinh z	$-\infty$	0	∞
cosh z	∞	1	∞
tanh z	-1	0	1

Note: $\sinh(-z) = -\sinh z$, $\cosh(-z) = \cosh z$

1.10.3 Relation between Circular and Hyperbolic Functions

(i) $\sin iz = i \sinh z$ and $\sinh z = -i \sin iz$

Proof : By Euler's formula,

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

Replacing z by iz,

$$\sin iz = \frac{e^{i^2 z} - e^{-i^2 z}}{2i} = -i\frac{\left(e^{-z} - e^{z}\right)}{2} \qquad \left[\because \frac{1}{i} = -i\right.$$

$$= i\frac{\left(e^{z} - e^{-z}\right)}{2} = i \sinh z$$

$$\sinh z = \frac{1}{i} \sin iz = -i \sin iz$$

(ii) $\cos iz = \cosh z$

Proof : By Euler's formula,

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

Replacing z by iz,

$$\cos iz = \frac{e^{i^2 z} + e^{-i^2 z}}{2} = \frac{e^{-z} + e^{z}}{2} = \cosh z$$

(iii) $\tan iz = i \tanh z$ and $\tanh z = -i \tan iz$

Proof :

$$\tan iz = \frac{\sin iz}{\cos iz} = \frac{i \sinh z}{\cosh z} = i \tanh z$$

$$\tanh z = \frac{1}{i} \tan iz = -i \tan iz$$

1.10.4 Formulae on Hyperbolic Functions

A. (i) $\cosh^2 z - \sinh^2 z = 1$

(ii) $\coth^2 z - \operatorname{cosech}^2 z = 1$

(iii) $\operatorname{sech}^2 z + \tanh^2 z = 1$

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B. (iv) $\sinh 2z = 2 \sinh z \cosh z$

(v) $\cosh 2z = \cosh^2 z + \sinh^2 z = 2\cosh^2 z - 1 = 1 + 2\sinh^2 z$

(vi) $\tanh 2z = \dfrac{2\tanh z}{1+\tanh^2 z}$

C. (vii) $\sinh 3z = 3\sinh z + 4\sinh^3 z$

(viii) $\cosh 3z = 4\cosh^3 z - 3\cosh z$

(ix) $\tanh 3z = \dfrac{3\tanh z + \tanh^3 z}{1+3\tanh^2 z}$

Proof : A. (i) For the circular functions, we have

$$\sin^2\theta + \cos^2\theta = 1$$

Putting $\theta = iz$,

$$(\sin iz)^2 + (\cos iz)^2 = 1$$
$$(i\sinh z)^2 + (\cosh z)^2 = 1$$
$$-\sinh^2 z + \cosh^2 z = 1$$
$$\cosh^2 z - \sinh^2 z = 1$$

Similarly, (ii) and (iii) can also be proved.

B. (iv) We have $\sin 2\theta = 2\sin\theta\cos\theta$

Putting $\theta = iz$,

$$\sin 2iz = 2\sin iz \cos iz$$
$$i\sinh 2z = 2i\sinh z\cosh z$$
$$\sinh 2z = 2\sinh z\cosh z$$

Similarly, (v) and (vi) can also be proved.

C. (ix) We have $\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1-\tan^2\theta}$

Putting $\theta = iz$,

$$\tan 3iz = \frac{3\tan iz - (\tan iz)^3}{1-3(\tan iz)^2}$$

$$i\tanh 3z = \frac{3i\tanh z - i^3\tanh^3 z}{1-3i^2\tanh^2 z}$$

$$\tanh 3z = \frac{3\tanh z + \tanh^3 z}{1+3\tanh^2 z}$$

Similarly, (vii) and (viii) can also be proved.

Similarly, we can prove the following formulae:

D. (x) $\sinh(z_1 \pm z_2) = \sinh z_1 \cosh z_2 \pm \cosh z_1 \sinh z_2$

(xi) $\cosh(z_1 \pm z_2) = \cosh z_1 \cosh z_2 \pm \sinh z_1 \sinh z_2$

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(xii) $\tanh(z_1 \pm z_2) = \dfrac{\tanh z_1 \pm \tanh z_2}{1 \pm \tanh z_1 \tanh z_2}$

E. (xiii) $\sinh z_1 + \sinh z_2 = 2 \sinh \dfrac{z_1 + z_2}{2} \cosh \dfrac{z_1 - z_2}{2}$

(xiv) $\sinh z_1 - \sinh z_2 = 2 \cosh \dfrac{z_1 + z_2}{2} \sinh \dfrac{z_1 - z_2}{2}$

(xv) $\cosh z_1 + \cosh z_2 = 2 \cosh \dfrac{z_1 + z_2}{2} \cosh \dfrac{z_1 - z_2}{2}$

(xvi) $\cosh z_1 - \cosh z_2 = 2 \sinh \dfrac{z_1 + z_2}{2} \sinh \dfrac{z_1 - z_2}{2}$

F. (xvii) $2 \sinh z_1 \cosh z_2 = \sinh(z_1 + z_2) + \sinh(z_1 - z_2)$

(xviii) $2 \cosh z_1 \sinh z_2 = \sinh(z_1 + z_2) - \sinh(z_1 - z_2)$

(xix) $2 \cosh z_1 \cosh z_2 = \cosh(z_1 + z_2) + \cosh(z_1 - z_2)$

(xviii) $2 \sinh z_1 \sinh z_2 = \cosh(z_1 + z_2) - \cosh(z_1 - z_2)$

1.11 INVERSE HYPERBOLIC FUNCTIONS

If $x = \sinh u$, then $u = \sinh^{-1} x$ is called sine hyperbolic inverse of x, where x is real. Similarly, we can define $\cosh^{-1} x$, $\tanh^{-1} x$, $\coth^{-1} x$, $\operatorname{sech}^{-1} x$ and $\operatorname{cosech}^{-1} x$.

The inverse hyperbolic functions are many valued functions but we will consider their principal values only.

If x is real,

(i) $\sinh^{-1} x = \log\left(x + \sqrt{x^2 + 1}\right)$ (ii) $\cosh^{-1} x = \log\left(x + \sqrt{x^2 - 1}\right)$

(iii) $\tanh^{-1} x = \dfrac{1}{2} \log\left(\dfrac{1+x}{1-x}\right)$

Proof : (i) Let $\sinh^{-1} x = y$

$$x = \sinh y = \frac{e^y - e^{-y}}{2}$$

$$2x = e^y - \frac{1}{e^y} = \frac{e^{2y} - 1}{e^y}$$

$$e^{2y} - 2x\, e^y - 1 = 0$$

This equation is quadratic in e^y.

$$e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2}$$

$$e^y = x \pm \sqrt{x^2 + 1}$$

$$y = \log\left(x \pm \sqrt{x^2 + 1}\right)$$

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But $x-\sqrt{x^2+1}<0$ and log negative is not defined.

$$y=\log\left(x+\sqrt{x^2+1}\right)$$

$$\sinh^{-1}x=\log\left(x+\sqrt{x^2+1}\right)$$

(ii) Let $\cosh^{-1}x=y$

$$x=\cosh y=\frac{e^y+e^{-y}}{2}$$

$$2x=e^y+\frac{1}{e^y}=\frac{e^{2y}+1}{e^y}$$

$$2xe^y=e^{2y}+1$$

$$e^{2y}-2xe^y+1=0$$

$$e^y=\frac{2x\pm\sqrt{4x^2-4}}{2}$$

$$e^y=x\pm\sqrt{x^2-1}$$

$$y=\log\left(x\pm\sqrt{x^2-1}\right) \quad \ldots (1)$$

Consider, $y=\log\left(x-\sqrt{x^2-1}\right)$, $e^y=x-\sqrt{x^2-1}$ ... (2)

$$e^{-y}=\frac{1}{x-\sqrt{x^2-1}}\cdot\frac{x+\sqrt{x^2-1}}{x+\sqrt{x^2-1}}=\frac{x+\sqrt{x^2-1}}{x^2-x^2+1}=x+\sqrt{x^2-1}$$

$$-y=\log\left(x+\sqrt{x^2-1}\right)$$

$$y=-\log\left(x+\sqrt{x^2-1}\right) \quad \ldots (3)$$

Equating Eqs. (2) and (3),

$$\log\left(x-\sqrt{x^2-1}\right)=-\log\left(x+\sqrt{x^2-1}\right) \quad \ldots (4)$$

From Eq. (1),

$$y=\pm\log\left(x+\sqrt{x^2-1}\right) \quad \ldots (5)$$

$$\cosh^{-1}x=\pm\log\left(x+\sqrt{x^2+1}\right)$$

$$x=\cosh\left[\pm\log\left(x+\sqrt{x^2+1}\right)\right]=\cosh\left[\log\left(x+\sqrt{x^2+1}\right)\right] \quad [\because \cosh(-z)=\cosh z]$$

$$\cosh^{-1}x=\log\left(x+\sqrt{x^2+1}\right)$$

(iii) Let $\tanh^{-1}x=y$

$$x=\tanh y$$

$$\frac{x}{1}=\frac{e^y-e^{-y}}{e^y+e^{-y}}$$

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Using componendo-dividendo,

$$\frac{1+x}{1-x}=\frac{e^{y}+e^{-y}+e^{y}-e^{-y}}{e^{y}+e^{-y}-e^{y}+e^{-y}}=\frac{2e^{y}}{2e^{-y}}=e^{2y}$$

$$e^{2y}=\frac{1+x}{1-x}$$

$$2y=\log\left(\frac{1+x}{1-x}\right)$$

$$y=\frac{1}{2}\log\left(\frac{1+x}{1-x}\right)$$

$$\tanh^{-1}x=\frac{1}{2}\log\left(\frac{1+x}{1-x}\right)$$

Example 1: Prove that $\left(\frac{1+\tanh x}{1-\tanh x}\right)^{3}=\cosh 6x+\sinh 6x$.

Solution: $\left(\frac{1+\tanh x}{1-\tanh x}\right)^{3}=\left(\frac{1+\frac{\sinh x}{\cosh x}}{1-\frac{\sinh x}{\cosh x}}\right)^{3}=\left(\frac{\cosh x+\sinh x}{\cosh x-\sinh x}\right)^{3}=\left(\frac{\cos ix-i\sin ix}{\cos ix+i\sin ix}\right)^{3}$

$$=\left[\left(\frac{\cos ix-i\sin ix}{\cos ix+i\sin ix}\right)\left(\frac{\cos ix-i\sin ix}{\cos ix-i\sin ix}\right)\right]^{3}$$

$$=\left[\frac{(\cos ix-i\sin ix)^{2}}{\cos^{2}ix+\sin^{2}ix}\right]^{3}$$

$$=(\cos ix-i\sin ix)^{6}$$

$$=\cos 6ix-i\sin 6ix \qquad \text{[Using De Moivre's theorem]}$$

$$=\cosh 6x-i\cdot i\sinh 6x$$

$$=\cosh 6x+\sinh 6x.$$

Example 2: Prove that $\dfrac{1}{1-\dfrac{1}{1-\dfrac{1}{1-\cosh^{2}x}}}=\cosh^{2}x$.

Solution: $\dfrac{1}{1-\dfrac{1}{1-\dfrac{1}{1-\cosh^{2}x}}}=\dfrac{1}{1-\dfrac{1}{1-\dfrac{1}{-\sinh^{2}x}}}$ $[\because \cosh^{2}x-\sinh^{2}x=1]$

$$=\frac{1}{1-\frac{1}{1+\operatorname{cosech}^{2}x}}$$

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$$= \frac{1}{1-\frac{1}{\coth^2 x}} \qquad [\because 1-\coth^2 x = -\operatorname{cosech}^2 x]$$

$$= \frac{1}{1-\tanh^2 x}$$

$$= \frac{1}{\operatorname{sech}^2 x} \qquad [\because 1-\tanh^2 x = \operatorname{sech}^2 x]$$

$$= \cosh^2 x.$$

Example 3: Prove that $\cosh^5 x = \frac{1}{16}[\cosh 5x + 5\cosh 3x + 10\cosh x]$.

Solution: $\cosh^5 x = \left(\frac{e^x + e^{-x}}{2}\right)^5$

$$= \frac{1}{2^5}(e^{5x} + 5e^{4x}\cdot e^{-x} + 10e^{3x}\cdot e^{-2x} + 10e^{2x}\cdot e^{-3x} + 5e^{x}\cdot e^{-4x} + e^{-5x})$$

$$= \frac{1}{2^5}\left[(e^{5x} + e^{-5x}) + 5(e^{3x} + e^{-3x}) + 10(e^{x} + e^{-x})\right]$$

$$= \frac{1}{2^5}(2\cosh 5x + 10\cosh 3x + 20\cosh x)$$

$$= \frac{1}{16}(\cosh 5x + 5\cosh 3x + 10\cosh x)$$

Example 4: Prove that $\tanh(\log\sqrt{3}) = 0.5$.

Solution: $\tanh(\log\sqrt{3}) = \frac{e^{\log\sqrt{3}} - e^{-\log\sqrt{3}}}{e^{\log\sqrt{3}} + e^{-\log\sqrt{3}}} = \frac{\sqrt{3} - \frac{1}{\sqrt{3}}}{\sqrt{3} + \frac{1}{\sqrt{3}}} = \frac{3-1}{3+1} = 0.5$

Example 5: Solve the equation $17\cosh x + 18\sinh x = 1$ for real values of x.

Solution: $17\cosh x + 18\sinh x = 1$

$$17\left(\frac{e^x + e^{-x}}{2}\right) + 18\left(\frac{e^x - e^{-x}}{2}\right) = 1$$

$$35e^x - e^{-x} = 2$$

$$35e^{2x} - 1 = 2e^x$$

$$35e^{2x} - 2e^x - 1 = 0$$

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This equation is quadratic in e^x.

$$e^x = \frac{2 \pm \sqrt{4+140}}{70} = \frac{2 \pm 12}{70} = \frac{14}{70}, \frac{-10}{70}$$

For real value, e^x should be positive.

$$e^x = \frac{1}{5}, \; x = \log \frac{1}{5} = -\log 5.$$

Example 6: Find $\tanh x$, if $\sinh x - \cosh x = 5$.

Solution: $\sinh x - \cosh x = 5$

$$\frac{e^x - e^{-x}}{2} - \frac{e^x + e^{-x}}{2} = 5$$

$$-e^{-x} = 5, \; e^{-x} = -5, \; e^x = \frac{-1}{5}$$

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{-\frac{1}{5} - (-5)}{-\frac{1}{5} + (-5)} = \frac{-1+25}{-1-25}$$

$$= \frac{24}{-26} = -\frac{12}{13}$$

Example 7: If $\cos\alpha\cosh\beta = \frac{x}{2}$, $\sin\alpha\sinh\beta = \frac{y}{2}$, prove that

(i) $\sec(\alpha - i\beta) + \sec(\alpha + i\beta) = \frac{4x}{x^2+y^2}$ (ii) $\sec(\alpha - i\beta) - \sec(\alpha + i\beta) = -\frac{4iy}{x^2+y^2}$.

Solution: $\sec(\alpha - i\beta) = \frac{1}{\cos(\alpha - i\beta)} = \frac{1}{\cos\alpha\cos i\beta + \sin\alpha\sin i\beta}$

$$= \frac{1}{\cos\alpha\cosh\beta + i\sin\alpha\sinh\beta}$$

$$= \frac{1}{\frac{x}{2} + i\frac{y}{2}} = \frac{2}{x+iy}$$

$$= \frac{2}{(x+iy)} \cdot \frac{(x-iy)}{(x-iy)} = \frac{2(x-iy)}{x^2+y^2} \quad \ldots (1)$$

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Similarly, $$\sec(\alpha+i\beta)=\frac{1}{\cos(\alpha+i\beta)}=\frac{1}{\cos\alpha\cosh\beta-i\sin\alpha\sinh\beta}$$

$$=\frac{1}{\frac{x}{2}-i\frac{y}{2}}=\frac{2}{x-iy}=\frac{2}{x-iy}\cdot\frac{(x+iy)}{(x+iy)}$$

$$=\frac{2(x+iy)}{x^2+y^2} \quad \ldots (2)$$

Adding Eqs. (1) and (2),

$$\sec(\alpha-i\beta)+\sec(\alpha+i\beta)=\frac{4x}{x^2+y^2}$$

Subtracting Eq. (2) from Eq. (1),

$$\sec(\alpha-i\beta)-\sec(\alpha+i\beta)=-\frac{4iy}{x^2+y^2}$$

Example 8: If $\log(\tan x)=y$, prove that

(i) $\cosh ny=\frac{1}{2}(\tan^n x+\cot^n x)$ (ii) $\sinh ny=\frac{1}{2}(\tan^n x-\cot^n x)$

(iii) $\sinh(n+1)y+\sinh(n-1)y=2\sinh ny\ \text{cosec}\ 2x$

(iv) $\cosh(n+1)y+\cosh(n-1)y=2\cosh ny\ \text{cosec}\ 2x$.

Solution: (i) $$\left.\begin{aligned}e^{y}&=\tan x\\ e^{-y}&=\cot x\end{aligned}\right\} \quad \ldots (1)$$

$$\cosh ny=\frac{e^{ny}+e^{-ny}}{2}=\frac{1}{2}(\tan^n x+\cot^n x)$$

(ii) $$\sinh ny=\frac{e^{ny}-e^{-ny}}{2}=\frac{1}{2}(\tan^n x-\cot^n x)$$

(iii) $$\sinh(n+1)y+\sinh(n-1)y=2\sinh\frac{(n+1+n-1)y}{2}\cosh\frac{(n+1-n+1)y}{2}$$

$$=2\sinh ny\cosh y=2\sinh ny\left(\frac{e^{y}+e^{-y}}{2}\right)$$

$$=2\sinh ny\left(\frac{\tan x+\cot x}{2}\right)=2\sinh ny\left(\frac{\sin^2 x+\cos^2 x}{2\sin x\cos x}\right)$$

$$=2\sinh ny\left(\frac{1}{\sin 2x}\right)=2\sinh ny\,\text{cosec}\,2x$$

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(iv) $\cosh(n+1)y+\cosh(n-1)y = 2\cosh\dfrac{(n+1+n-1)y}{2}\cosh\dfrac{(n+1-n+1)y}{2}$

$= 2\cosh ny\cosh y = 2\cosh ny\left(\dfrac{e^{y}+e^{-y}}{2}\right)$

$= 2\cosh ny\left(\dfrac{\tan x+\cot x}{2}\right) = 2\cosh ny\left(\dfrac{\sin^2 x+\cos^2}{2\sin x\cos}\right.$

$= 2\cosh ny\operatorname{cosec} 2x$

Example 9: If $u = \log\left[\tan\left(\dfrac{\pi}{4}+\dfrac{\theta}{2}\right)\right]$, prove that

(i) $\tanh\dfrac{u}{2} = \tan\dfrac{\theta}{2}$ **(ii) $\cosh u = \sec\theta$.**

Solution: (i) $u = \log\left[\tan\left(\dfrac{\pi}{4}+\dfrac{\theta}{2}\right)\right]$

$$e^{u} = \frac{\tan\frac{\pi}{4}+\tan\frac{\theta}{2}}{1-\tan\frac{\pi}{4}\tan\frac{\theta}{2}}$$

$$\frac{e^{u}}{1} = \frac{1+\tan\frac{\theta}{2}}{1-\tan\frac{\theta}{2}}$$

$$u = \log\left(\frac{1+\tan\frac{\theta}{2}}{1-\tan\frac{\theta}{2}}\right) = 2\tanh^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$\frac{u}{2} = \tanh^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$\tanh\frac{u}{2} = \tan\frac{\theta}{2}$$

(ii) From part (i),

$$\tanh\frac{u}{2} = \tan\frac{\theta}{2}$$

$$\tanh^2\frac{u}{2} = \tan^2\frac{\theta}{2}$$

Using componendo-dividendo,

$$\frac{1+\tanh^2\frac{u}{2}}{1-\tanh^2\frac{u}{2}} = \frac{1+\tan^2\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}}$$

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$$\cosh u = \sec\theta \quad \left[\because \cos\theta = \frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}} \text{ and } \cosh u = \frac{1+\tanh^2\frac{u}{2}}{1-\tanh^2\frac{u}{2}}\right]$$

Example 10: Prove that $\sinh^{-1}(\tan\theta) = \log\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)$.

Solution: $\sinh^{-1}(\tan\theta) = \log\left(\tan\theta+\sqrt{\tan^2\theta+1}\right)$

$$= \log(\tan\theta+\sec\theta) = \log\left(\frac{\sin\theta+1}{\cos\theta}\right)$$

$$= \log\left[\frac{\cos\left(\frac{\pi}{2}-\theta\right)+1}{\sin\left(\frac{\pi}{2}-\theta\right)}\right] = \log\left[\frac{2\cos^2\left(\frac{\pi}{4}-\frac{\theta}{2}\right)}{2\sin\left(\frac{\pi}{4}-\frac{\theta}{2}\right)\cos\left(\frac{\pi}{4}-\frac{\theta}{2}\right)}\right]$$

$$= \log\left[\cot\left(\frac{\pi}{4}-\frac{\theta}{2}\right)\right] = \log\left[\tan\left(\frac{\pi}{2}-\frac{\pi}{4}+\frac{\theta}{2}\right)\right]$$

$$= \log\left[\tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)\right].$$

Example 11: Prove that

(i) $\tanh^{-1}x = \sinh^{-1}\frac{x}{\sqrt{1-x^2}}$ **(ii) $\sinh^{-1}x = \cosh^{-1}\left(\sqrt{1+x^2}\right)$.**

Solution:

(i) $\sinh^{-1}\frac{x}{\sqrt{1-x^2}} = \log\left(\frac{x}{\sqrt{1-x^2}}+\sqrt{\frac{x^2}{1-x^2}+1}\right)$ $\left[\because \sinh^{-1}x = \log\left(x+\sqrt{x^2+1}\right)\right]$

$$= \log\left(\frac{x}{\sqrt{1-x^2}}+\frac{1}{\sqrt{1-x^2}}\right) = \log\frac{x+1}{\sqrt{1-x^2}}$$

$$= \log\frac{\sqrt{1+x}\sqrt{1+x}}{\sqrt{1-x}\sqrt{1+x}} = \log\frac{\sqrt{1+x}}{\sqrt{1-x}} = \frac{1}{2}\log\frac{1+x}{1-x} = \tanh^{-1}x.$$

(ii) $\cosh^{-1}\left(\sqrt{1+x^2}\right) = \log\left(\sqrt{1+x^2}+\sqrt{1+x^2-1}\right)$ $\left[\because \cosh^{-1}x = \log\left(x+\sqrt{x^2-1}\right)\right]$

$$= \log\left(\sqrt{1+x^2}+x\right) = \sinh^{-1}x.$$

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